

ABDELRAHMAN MOHAMED



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AbdelrahmanAyman0011

EDUCATION

Alexandria National University

graduated June 2026

Bachelor of Computer Science - **Cybersecurity** Department

Alexandria, Egypt

- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Computer Networks, Network Security, Data Science, Social Networks Computing, Cryptography, Distributed Systems, AI & ML.

EXPERIENCE

Early-Stage Startup | System Engineer

Oct 2025 – Present | Remote - USA

- Architected end-to-end **AWS cloud infrastructure** for high **availability** and reliability, prioritizing resilient system design and low operational costs; automated resource provisioning across services using **Terraform (IaC)**.
- Engineered a high-throughput, scalable parallel data pipeline using **PostgreSQL** as an event producer, designing custom **UDF-based batching mechanisms** to dispatch streaming records across concurrent background workers.
- Refactored legacy **User-Defined Functions (UDFs)** into polyglot, event-driven serverless services across **Rust, Python,** and **Java** running on **AWS Lambda**.
- Built isolated local cloud emulation suites with **LocalStack** and **Floci**, accelerating team feedback loops, offline debugging, and comprehensive Lambda regression testing.
- Improved **performance** and analytics turnaround across massive datasets by optimizing queries and partitioned storage using **Amazon Athena** and advanced **SQL**.

Pixelogic Media | Devops Intern

Jul 2026 – Sept 2026 | Cairo, Egypt

- Transitioned cloud infrastructure management to **Infrastructure as Code (IaC)** using **Terraform** and **Terragrunt**, architecting and provisioning the entire **AWS IAM** service layer.
- Engineered automated refactoring and validation scripts to overhaul the entire company's tagging architecture, enabling granular **cost analysis** and accurate budget management across all individual teams and projects.
- Monitored production infrastructure health, troubleshooted operational issues, and shadowed senior engineers to streamline cloud operations, system reliability, and deployment workflows.

Meta | Network Production Engineering Intern

June 2025 – Sept 2025 | Dublin, Ireland

- Contributed to Meta's network infrastructure team, supporting **large-scale production network reliability**.
- Worked on **network policy verification** to validate infrastructure-wide configuration changes against established routing policies, ensuring changes across production systems did not violate network constraints.
- Developed **Python automation** tools to monitor network operations, troubleshoot issues, and detect potential failures.
- Used **Linux-based** systems to implement mitigation strategies and maintain network stability at scale.

Information Technology Institute (ITI) | Linux Intern

July 2024 – August 2024 | Remote

- Focused on Linux administration, Red Hat operating systems, managing infrastructure, and virtualization.
- Gained hands-on experience with **Linux/UNIX systems**, managing users, monitoring, services, and system security.

PROJECTS

K-rypta (Graduation Project) | Python, WebRTC, Cryptography, P2P Architecture

- Developed a secure, local-first **decentralized password manager** that eliminates cloud dependencies to minimize attack surfaces, prioritizing data sovereignty and user privacy.
- Implemented robust cryptographic primitives including **AES-GCM** authenticated encryption; designed a hybrid P2P architecture using **WebRTC** for encrypted device-to-device synchronization.

Secure Distributed System Observability | Python, RabbitMQ, PostgreSQL, Docker, Nginx

- Architected a production-grade containerized system with a full observability pipeline (Nginx → API → RabbitMQ → Worker); implemented a **UUID correlation system** for end-to-end request tracing.
- Developed a **Security Education Suite** featuring a Man-in-the-Middle (MITM) demo and automated certificate generation scripts to visualize the impact of TLS/HTTPS termination.

Network Simulation | Cisco Packet Tracer, Networking, Subnetting

- Implemented LAN topologies using Cisco Packet Tracer for efficient network design and connectivity, configured IP addressing and subnetting schemes, and collaborated on solutions for network segmentation and management.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, PHP, C

Networking & Systems: Linux/UNIX, Kernel concepts, TCP/IP, UDP, ICMP, WebRTC, P2P, DNS, Routing Protocols

Tools & Technologies: Git, GitHub, Docker, AWS Lambda, S3, EKS, ECR, Athena, PostgreSQL, Load Balancing

Concepts: OOP, Data Structures & Algorithms, Networking, Troubleshooting, Distributed Systems

ACHIEVEMENTS

Achieved **first place** in university's **ECPC** qualifications and advanced to the ECPC competition for **two consecutive years**.

AWS Academy Graduate: Cloud Security Foundations (May 2025) & Cloud Foundations (Dec 2024).

Led problem-solving training sessions, mentoring peers in programming and algorithms.